

deConfUSlon fUSI Studio

Full Updated User Manual

Source-reviewed edition based on the current toolbox ZIP, the 09.06.2026 manual, and the 24.06.2026 Standardized Analysis update

Item	Value
Manual version	2026-06-24 - full consolidated update
Toolbox name	deConfUSlon / fUSI Studio
Legacy name	HUMoR / HUMOR-Analysis-Tool
Main launcher	deConfUSlon.m -> run_fusi_studio.m
Current source reviewed	deConfUSlon.zip uploaded 2026-06-24; static code review of active files, excluding .git/backups for runtime inventory
Primary scope	2D fUSI, step-motor / multi-slice fUSI, PSC/SCM, masks, atlas registration, segmentation, FC, Group Analysis
Important limitation	The source was statically inspected for this manual. MATLAB runtime validation on representative datasets is still required before publication use.
Research-use warning deConfUSlon is research software, not medical software. Always manually review raw movies, QC figures, preprocessing choices, baseline windows, masks, atlas registration, ROI traces, FC matrices, group labels, and exported figures before interpretation or publication.	

Table of contents

1. 1. Purpose and scope
 2. 2. Abbreviations and key terms
 3. 3. Core mathematical and imaging principles
 4. 4. Installation and starting the toolbox
 5. 5. Data types, dimensions, and folder organization
 6. 6. Main Studio window and module map
 7. 7. Standardized Analysis: workflow A, workflow B, editing, save/load
 8. 8. Recommended individual-subject workflow
 9. 9. Quality-control workflow
 10. 10. Preprocessing modules and decision logic
 11. 11. Frame Rejection vs Scrubbing vs Despiking
 12. 12. Visualization: Time-Course Viewer, Video GUI, SCM GUI
 13. 13. Mask Editor
 14. 14. Atlas registration and segmentation
 15. 15. Functional Connectivity
 16. 16. Group Analysis and FC Group Analysis
 17. 17. Outputs, naming, bundles, and logs
 18. 18. Publication-ready validation checklist
 19. 19. Troubleshooting
 20. 20. Source-code orientation and release notes
- Appendix A. Formula quick reference
 - Appendix B. Active source-file inventory
 - Appendix C. Change log for this manual update

1. Purpose and scope

This manual explains how to use deConfUSlon / fUSI Studio for functional ultrasound imaging analysis, from loading data to QC, preprocessing, PSC/SCM visualization, masking, atlas registration, segmentation, functional connectivity, and group-level export.

It consolidates the older 09.06.2026 manual structure with the 24.06.2026 Standardized Analysis update and the current source code orientation. The focus is practical use: what each module does, when to use it, what to check, and how to avoid common mistakes.

Core philosophy

Do not treat the GUI as a black box. Every processing step changes the active dataset. The safe workflow is: load raw data, verify dimensions/TR, QC, apply only justified corrections, inspect the result, then proceed to atlas, segmentation, FC, and group analysis.

1.1 What this manual covers

- Launching the current split-GUI architecture using deConfUSlon.m and run_fusi_studio.m.
- 2D [Y X T] and step-motor / multi-slice [Y X Z T] workflows.
- Workflow A Fast and Workflow B Detailed in Standardized Analysis.
- Editing workflow order, checking/unchecking steps, changing numeric presets, and saving/loading a custom preset.
- Quality control outputs and what they mean.
- Practical differences between Frame Rejection, Scrubbing, and Despiking.
- Mathematical background for PSC, baseline handling, temporal filtering, smoothing, tSNR, CNR, DVARS, and FC.
- SCM, Video GUI, Mask Editor, atlas registration, segmentation, FC, and Group Analysis.
- Current source-code orientation and release/cleanup recommendations.

1.2 What this manual does not replace

- MATLAB runtime testing on real 2D, step-motor, and atlas-registered datasets.
- Biological validation of fUSI signals, masks, atlas alignment, or functional connectivity interpretation.
- Local machine-specific setup for hardware acquisition, serial ports, PowerPoint automation, or ActiveX export.

2. Abbreviations and key terms

Term	Meaning in deConfUSlon
deConfUSlon	Current user-facing toolbox name and active repository name. Some helper prefixes use deConfUSlon_.
HUMoR / HUMOR	Legacy project/toolbox name that may remain in comments, metadata, old outputs, or compatibility helper fields.
fUSI	Functional ultrasound imaging.
PD / Power Doppler	Power Doppler-like signal intensity used as the fUSI signal source.
CBV	Cerebral blood volume. fUSI power Doppler is often interpreted as CBV-like hemodynamic signal.
TR	Time between frames or volumes, in seconds.
PSC	Percentage signal change: $100 * (\text{signal} - \text{baseline}) / \text{baseline}$.
SCM	Signal-change map; spatial map of PSC or related response values.
ROI	Region of interest.
QC	Quality control.
tSNR	Temporal signal-to-noise ratio; mean signal over time divided by temporal standard deviation.
CNR	Contrast-to-noise ratio; signal difference between windows divided by noise estimate.
DVARS	Framewise root-mean-square temporal derivative metric used to detect abrupt global changes.
COM	Center of mass; used for apparent drift or motion QC.
PCA / ICA	Principal component analysis / independent component analysis; optional component-based denoising.
FC	Functional connectivity, usually correlation between seed/ROI time courses.
Fisher z	$\text{atanh}(r)$ transform used to average and test correlation coefficients.
Allen atlas / CCF	Allen Common Coordinate Framework atlas resources used for

	anatomical registration/labels.
Reg2D	Saved 2D coronal registration object for 2D or step-motor atlas workflows.
JM order/color	Atlas color/order support using atlas_tools/rgb2acr.xlsx and list_selected_regions.txt.
Underlay	Background/anatomical image used beneath overlays in SCM, Video, FC, or Group Analysis.
Overlay / signal mask	Mask restricting the displayed or analyzed signal region.
Brain / underlay mask	Mask restricting valid anatomy/background region.
Bundle	MAT export containing data plus metadata needed to reopen or attach results in SCM, Video, FC, or Group Analysis.
Workflow A	Fast standardized review workflow.
Workflow B	Detailed standardized subject-level workflow.

3. Core mathematical and imaging principles

3.1 fUSI signal interpretation

The fUSI signal in this toolbox is treated as a Power-Doppler-like intensity time series. It is commonly interpreted as a hemodynamic or CBV-like signal, not direct neuronal firing. Therefore, motion, bubbles, skull shadows, probe changes, baseline drift, and non-brain voxels can create apparent responses or false functional connectivity.

Interpretation rule

A strong map is not automatically a biological response. It must be stable in the movie, compatible with the baseline and timing, inside a valid mask, and reproducible across validated subjects.

3.2 Data dimensions

Data type	Typical dimensions	Meaning
2D / single-slice fUSI	[Y X T]	Y and X are spatial axes; T is time.
Step-motor / multi-slice fUSI	[Y X Z T]	Z is slice index; T is time after reconstruction.
Matrix / 3D-like data	[Y X Z T]	Z is a true spatial dimension or volume index depending on acquisition.
Static image / underlay	[Y X] or [Y X Z]	Used for display, atlas registration, masks, or mean images.

3.3 Baseline and PSC

The baseline window is the reference period used to convert raw intensity to percentage signal change. A bad baseline contaminates every PSC frame and every SCM map derived from it.

$$B(x,y,z) = \text{mean}_t I(x,y,z,t), \text{ for } t \text{ in baseline frames}$$

$$\text{PSC}(x,y,z,t) = 100 * (I(x,y,z,t) - B(x,y,z)) / B(x,y,z)$$

- Use a stable pre-stimulus/pre-injection baseline whenever possible.
- Do not include injection artifacts, motor movement, transient settling, or obvious motion in baseline.
- For step-motor data, check that the baseline corresponds to the same reconstructed slice timing logic used downstream.
- Always report baseline start/end in seconds and ensure the same units are used in SCM, Video, and group export.

3.4 Temporal interpolation and effective TR

computePSC.m supports temporal interpolation with an integer factor N. Interpolation increases internal samples but should preserve the total experiment duration by using an effective TR.

$$n\text{Frames_interpolated} = (n\text{Frames_original} - 1) * N + 1$$

$$\text{TR_effective} = \text{TR_original} / N$$

$$\text{Tmax} = (n\text{Frames_interpolated} - 1) * \text{TR_effective}$$

3.5 Temporal smoothing and subsampling

Temporal smoothing reduces high-frequency noise by averaging nearby frames. Subsampling/block averaging reduces the number of frames and changes the effective TR.

Sliding smoothing: output length = input length
 Block subsampling: output TR = original TR * block_size_frames
 block_size_frames = max(1, round(window_seconds / TR))

3.6 Temporal filtering

filtering.m applies low-pass, high-pass, or band-pass Butterworth filtering along the time dimension. The current implementation supports [Y X T] and [Y X Z T], returns static 2D images unchanged, and restores the voxelwise mean by default so the filtered output does not look artificially dark.

Nyquist frequency = $1 / (2 * TR)$
 Band-pass example: keep frequencies between FcLow and FcHigh
 Default standardized editable fields: filterType, FcLow, FcHigh, filterOrder

3.7 QC metrics

Metric	Calculation idea	Use
tSNR	$tSNR = \text{mean}(\text{signal over time}) / \text{std}(\text{signal over time})$	Low tSNR can indicate noisy tissue, poor coupling, bad mask, or motion.
CNR	$CNR \text{ approximately } = \text{abs}(\text{mean}(\text{signal window}) - \text{mean}(\text{baseline})) / \text{std}(\text{baseline})$	Requires sensible baseline and signal windows.
DVARs	$DVARs(t) = \text{RMS over voxels of } I(t) - I(t-1)$	Large values suggest abrupt framewise changes.
Common-mode portion	Portion of signal explained by shared/global temporal structure	High common-mode can mean global artifact dominance.
COM drift	Change in intensity-weighted center of mass over time	Apparent motion/drift diagnostic, not a true rigid-body transform.

3.8 Functional connectivity and Fisher z

Functional connectivity is usually computed as Pearson correlation between seed or ROI time courses. Correlations should be averaged and statistically compared in Fisher z-space, then converted back to Pearson r for display if needed.

```
r = corr(x, y)
Fisher_z = atanh(r)
mean_r_for_display = tanh(mean(Fisher_z_values))
```

4. Installation and starting the toolbox

4.1 Recommended MATLAB launch

Open MATLAB and change to the toolbox root. For the current repository name, use:

```
cd('D:\Github\deConfUSIon')
deConfUSIon
```

deConfUSIon.m is the user-facing launcher. It adds the toolbox root and atlas_tools to the MATLAB path and calls run_fusi_studio.m. run_fusi_studio.m assembles fusi_studio_GUI.m and fusi_studio_callback.m into the runtime GUI.

Do not remove run_fusi_studio.m

The split GUI depends on run_fusi_studio.m. Older instructions that mention fusi_studio.m are outdated for the current split-GUI source.

4.2 Platform notes

- Main development/testing context: Windows, MATLAB 2017b compatibility style, active use in MATLAB R2023b.
- Some acquisition scripts use local serial-port names and must be edited for each workstation.
- Some PowerPoint exports may require Microsoft PowerPoint / ActiveX support on Windows.
- Backups and .git should not be added recursively to the MATLAB path during normal use.

5. Data types, dimensions, and folder organization

5.1 Supported inputs

Input	Expected content	Important check
.mat	Image variable such as I, or supported nested fields; [Y X T] or [Y X Z T].	Verify variable, dimensions, TR, and metadata.

.nii	NIfTI image stack.	Verify orientation, dimensions, time axis, and TR manually.
.nii.gz	Compressed NIfTI read via gunzip/read workflow where supported.	Same checks as NIfTI.
Step-motor split MAT folder	Many slice/time MAT files, usually slice*_t*.mat style.	Use Motor reconstruction and verify slice count/order.
SCM/FC/Group bundles	MAT files exported from SCM GUI, Video GUI, Functional Connectivity, or Group Analysis.	Use the matching GUI/workflow; do not confuse static map exports with full bundles.

5.2 Recommended folder structure

```

RawData/
  project/animal/session/scan.mat
AnalysedData/
  project/animal/session/scan/
    QC/
    Preprocessing/
    PSC/
    Visualization/
    Masks/
    Registration/
    Registration2D/
    Segmentation/
    FunctionalConnectivity/
    GroupAnalysis/
    Reports/

```

Raw data rule

Never overwrite raw data. Each preprocessing step should create a new dataset version. Always confirm the active dataset dropdown before SCM, Video, Registration, Segmentation, FC, or Group Analysis.

6. Main Studio window and module map

The current fusi_studio_GUI.m defines ten visible sections. Sections 1-7 are the main validated user workflow. Sections 8-10 are visible but should be treated as placeholders/reserved unless the callbacks are fully validated.

Studio section	Buttons	Purpose / status
1. Dataset	Load fUSI Data	Load MAT/NIfTI data and choose the active dataset.
2. QC & Data Overview	Full QC, Specific QC	Run diagnostics before and after preprocessing.
3. Recommended Processing	Frame Rejection, Imregdemons, Scrubbing, Motor	Correct known acquisition artifacts or reconstruct step-motor data.
4. Advanced Processing	Temporal Smoothing/Subsampling, Filtering, PCA / ICA, Despike	Optional denoising; only use with documented rationale.
5. Visualization	Time-Course Viewer, SCM GUI, Video GUI, Mask Editor	Inspect raw/PSC dynamics, maps, videos, masks, and underlays.
6. Coregistration	Registration to Atlas, Segmentation	Align data to atlas and extract atlas-region time courses.
7. Advanced Analysis	Functional connectivity, Group analysis	Subject-level FC and cohort-level analysis/export.
8. GLM / Regression	General Linear Models, Regression	Placeholder/reserved in current source.
9. Velocity Analysis	Velocity Maps, Flow / Velocity QC	Placeholder/reserved in current source.
10. Community Analysis	Standardized Analysis, Placeholder Community B	Standardized Analysis is active; Placeholder Community B is reserved.

7. Standardized Analysis: workflow A, workflow B, editing, save/load

Standardized Analysis is the guided workflow manager. It does not replace manual review. It reduces repeated setup work by launching selected Studio modules in a controlled order and passing standardized numeric presets where supported.

7.1 How the Standardized Analysis window works

- Clicking a row only selects it; it does not activate or deactivate the step.
- The Run checkbox decides whether the row runs or is skipped.
- The displayed order controls execution order. You can edit the ORDER field or use Move Up, Move Down, Top, and Bottom.
- The Selected step settings panel shows editable numeric presets for the currently selected row.
- Run Workflow executes all ticked rows in the displayed order.
- Run Step executes only the currently selected step.
- Interactive steps open their own GUI and the workflow waits until that GUI is closed before continuing.
- Save / Load stores or restores the current workflow preset in standardizedAnalysis_preset.mat next to standardizedAnalysis.m.

Important run/selection rule

The checkbox controls Run/Skip. Row clicking only selects the row so you can inspect or edit its parameters. This prevents accidental workflow changes while browsing rows.

7.2 Supported editable numeric settings

Step	Editable fields	How to use them
Motor	Slices	Default 4; sanitized to 1-12.
Imregdemons	nsub / imreg n	Default 25; block size for median/mean block registration.
Filtering	Filter type, low cutoff, high cutoff, order	Type 1=band, 2=low, 3=high; order clamped to 1-6.
Temporal Smoothing	Mode, smoothing window, nsub, block method	Mode 1=sliding smoothing, 2=block/subsample; method 1=mean, 2=median.
PCA / ICA	Method, PCA max components, ICA components	Method 1=PCA, 2=ICA; component GUI still requires manual review.
Video GUI / SCM GUI	Baseline start/end, display min/max, alpha min/max	Defaults 30-35 s, caxis -100..100, alpha -20..20.
Time-Course Viewer	Baseline start/end, display min/max	Used for initial trace inspection.
Functional Connectivity	Display min/max	Used for FC display scaling, often -1..1.

7.3 Workflow A Fast: quick review path

Workflow A is intended for a rapid first-pass inspection. It checks whether the dataset can be reconstructed/corrected and whether video, time courses, and SCM maps look plausible before investing time in masks, atlas registration, segmentation, and FC.

Status	Step	Principle
Ticked	Motor	Step-motor/split-folder reconstruction; default 4 slices, residual despiking, correction none/raw.
Ticked	Imregdemons	Spatial correction using median mode and nsub 25; step-motor per-slice behavior.
Ticked	Video GUI	Baseline 30-35 s, signed PSC, caxis -100..100, alpha modulation -20..20.
Ticked	Time-Course Viewer	Inspect raw/PSC time courses and baseline stability.
Ticked	SCM GUI	Native-space SCM review; only the first SCM row is ticked.
Unticked	Everything else	Available but skipped: QC before/after, masks, atlas, segmentation, FC, Group Analysis, optional denoising.

Use Workflow A when

You want quick triage: Does the data load? Is slice reconstruction plausible? Does spatial correction help? Are there obvious time-course or map artifacts? Workflow A should not be used as the final publication workflow by itself.

7.4 Workflow B Detailed: complete subject-level path

Workflow B is the detailed default route for a subject that can proceed toward atlas registration, segmentation, FC, and group export. Optional denoising steps remain visible but unticked so the user must deliberately enable them.

Order	Step	Default	Reason
1	Motor	Ticked	Reconstruct or handle step-motor data first.
2	Full QC before preprocessing	Unticked by default	Available if you want pre-correction QC after Motor.
3	Frame Rejection	Unticked	Enable only if complete frames are rejected by QC.
4	Scrubbing	Unticked	Enable only for temporal outlier frames/DVARS/global signal artifacts.
5	Filtering	Unticked	Enable only with defined frequency rationale.
6	Temporal Smoothing	Unticked	Enable only if smoothing/subsampling is scientifically acceptable.
7	PCA / ICA	Unticked	Enable only after manual component review.
8	Despike	Unticked	Enable only for sparse voxel-wise spikes.
9	Imregdemons	Ticked	Spatial correction after optional early steps.
10	Full QC after preprocessing	Ticked	Check the corrected/processed output.
11	Mask Editor	Ticked	Create or load valid masks/underlays.
12	Video GUI	Ticked	Dynamic visual inspection.
13	Time-Course Viewer	Ticked	ROI/global trace inspection.
14	SCM GUI before atlas	Ticked	Native-space SCM map check.
15	Registration to Atlas	Ticked	Manual atlas alignment.
16	SCM GUI after atlas	Ticked	Re-check SCM with atlas-space context.
17	Segmentation	Ticked	Extract atlas-region time courses.
18	Functional Connectivity	Ticked	Compute subject FC after labels are validated.
19	Group Analysis	Unticked	Normally run after all subject-level exports are ready.

7.5 How to create your own workflow

21. Open Standardized Analysis from the main Studio window.
22. Choose Option A Fast or Option B Detailed as a starting point.
23. Click a row to select it and inspect the description/settings.
24. Check or uncheck Run for steps you want to include or skip.
25. Use Move Up, Move Down, Top, Bottom, or edit the ORDER field to change execution order.
26. Edit numeric settings in the Selected step settings panel where available.
27. Click Save / Load -> Save to store the current workflow as standardizedAnalysis_preset.mat.
28. Later, click Save / Load -> Load to restore your saved workflow.
29. Run Step to test one module; Run Workflow only after the sequence and settings are correct.

Preset safety

Save / Load stores the row structure and numeric settings. It does not prove the workflow is biologically correct. Always document why optional steps were enabled and verify outputs after each major processing stage.

8. Recommended individual-subject workflow

Step	Action
0. Prepare session	Confirm animal/session/scan, paradigm, TR, acquisition duration, stimulus/injection timing, baseline window, expected slices, and output folder.
1. Launch and load	Run deConfUSlon, load data, confirm active dataset label, probe type, dimensions, TR, and total duration.
2. Raw QC	Run Full QC or Specific QC, inspect raw movie, global trace, and frame issues.
3. Reconstruct / correct	Use Motor for step-motor data; use imregdemons only if spatial

	drift/motion justifies it.
4. Optional denoising	Use frame rejection, scrubbing, filtering, temporal smoothing, PCA/ICA, or despiking only with a specific QC reason.
5. Post-processing QC	Rerun QC after important corrections and compare to raw QC.
6. Visual inspection	Use Time-Course Viewer, Video GUI, and SCM GUI to inspect dynamics and PSC maps.
7. Masks and underlays	Create/load masks in Mask Editor; verify underlay and overlay match current data and slice.
8. Atlas registration	Register to atlas only after data/masks/underlay are stable; inspect every slice.
9. Segmentation	Extract atlas-region time courses only if registration and label maps are correct.
10. Functional Connectivity	Load/verify ROI labels and names, compute ROI current/seed results, inspect matrices and maps.
11. Export bundles	Export SCM/FC GroupAnalysis bundles after subject-level checks.
12. Group level	Load validated bundles into existing GroupAnalysis rows, recompute, inspect, and export.

9. Quality-control workflow

QC should be run before major preprocessing when possible and again after important correction steps. QC is diagnostic, not a final acceptance certificate.

QC module	Output	What to inspect
Frequency QC	Power spectra in broad and low-frequency bands.	Rhythmic acquisition noise, low-frequency drift, physiological or hardware periodicities.
Spatial QC	Mean image, temporal CV, tSNR map/histogram.	Vessel visibility, noisy regions, mask errors, poor coupling.
Temporal QC	Global signal, relative global signal, DVARS, spikes.	Abrupt jumps, drift, motion-related events.
Frame-rate QC	Intensity distribution and rejected-volume trace.	Complete-frame outliers and acquisition irregularities.
Stability QC	Intensity distribution, Gaussian/threshold style stability, rejection trace/map.	Stable acquisition support across frames/slices.
Motion / COM QC	Center-of-mass drift.	Apparent movement of signal mass over time.
Burst QC	Burst ratio, noisy voxel fraction, coverage over time.	Burst artifacts or acquisition instability.
CNR QC	True CNR if windows are provided; otherwise pseudo-CNR.	Response contrast relative to baseline variability.
Common-mode QC	Block correlation / common-mode portion.	Global artifacts dominating local signals.
Reliability QC	Finite/non-NaN support and optional atlas-region reliability.	Unreliable regions, bad slices, missing values.
PCA QC	Explained variance and first components.	Dominant artifact structure or unexpected components.

QC acceptance rule

A dataset can pass numeric QC and still be unsuitable if the raw movie, baseline, mask, registration, or ROI trace is wrong. Always use QC together with visual inspection.

10. Preprocessing modules and decision logic

10.1 Motor / step-motor reconstruction

motor.m reconstructs step-motor data into [Y X Z T]. It supports continuous mode and split-folder mode. In standardized mode it prefers split MAT folder behavior, no/raw correction, residual despiking enabled, spike threshold 4, and editable slice count default 4.

- Use Motor when the acquisition cycles through slices or saves split slice/time MAT files.
- Check output dimensions and slice count immediately after reconstruction.
- Confirm that later SCM, Video, FC, and GroupAnalysis modules preserve the Z slice dimension.
- If output looks wrong, inspect frame counts, naming, slice_t grouping, and acquisition log.

10.2 Imregdemons spatial correction

imregdemons_preprocess.m performs block reduction followed by non-rigid drift correction. It supports [Y X T] and [Y X Z T]. For step-motor data, per-slice 2D registration is preferred over treating all slices as one volume when the acquisition is slice-interleaved.

- Default standardized setting: median block method, nsub 25, QC saving enabled, QC display disabled during automated launch.
- Use when slow spatial drift or deformation affects downstream maps/ROI traces.
- Do not use blindly if the raw movie is stable; registration can introduce artifacts if settings are inappropriate.
- Inspect corrected output and QC before trusting SCM, segmentation, or FC.

10.3 Filtering

filtering.m applies Butterworth low/high/band-pass filtering to temporal fluctuations. It restores the voxelwise mean by default and handles [Y X T] and [Y X Z T].

Parameter	Meaning	Typical caution
Filter type	1=band-pass, 2=low-pass, 3=high-pass in Standardized Analysis.	Do not use a filter just because it improves the picture; define the biology/noise band.
FcLow	Lower cutoff frequency in Hz.	Too high removes slow biological responses.
FcHigh	Upper cutoff frequency in Hz.	Must be below Nyquist = $1/(2*TR)$.
Order	Butterworth order, clamped 1-6.	Higher order can create stronger edge/ringing behavior.
restoreMean	Current code default true.	Preserves mean image while filtering fluctuations.

10.4 Temporal smoothing / subsampling

temporalsmoothing.m has two modes: sliding smoothing and block averaging/subsampling. Sliding mode keeps the same number of frames; block mode reduces frames and changes the effective TR.

Mode	What it does	When to use	Risk
Sliding	Centered moving average with endpoint handling.	Noisy traces where temporal smoothing is acceptable.	Can blur fast responses.
Block/subsample	Non-overlapping mean or median blocks; output length reduced.	Long data where temporal resolution can be sacrificed.	Changes TR and may complicate stimulus timing.

10.5 PCA / ICA denoising

PCA/ICA opens an interactive component workflow. It can help remove artifact components but can also remove real biological responses if used carelessly. Standardized Analysis can preselect PCA or ICA and component limits, but user review remains essential.

- Inspect component spatial maps and time courses.
- Do not remove components only because they explain high variance.
- Record which components were removed and why.
- Compare SCM/time courses before and after PCA/ICA.

11. Frame Rejection vs Scrubbing vs Despike

These three modules overlap because all can address bad time points. The key difference is the unit of correction: complete frames, framewise temporal outlier volumes, or individual voxel spikes.

Module	Correction unit	Detection idea	Best use	Typical action	Main risk
Frame Rejection	Whole frames / volumes	Global intensity or frame-rate style outliers; rejected volumes over time.	Complete acquisition frames are bad or missing.	Usually interpolation/removal of rejected frames.	Can remove too much data if threshold is too strict.
Scrubbing	Whole frames / volumes, often based on temporal derivatives	DVARs or global signal; optional trimming; linear or PCHIP interpolation.	Motion/artifact frames are visible as abrupt temporal outliers.	Interpolates detected bad frames; can trim start/end.	Can distort temporal statistics if many frames are scrubbed.
Despike	Individual voxel-time points	Voxel-wise robust median/MAD z-score.	Sparse single-voxel or local spikes occur while most of the frame is usable.	Interpolates only spike time points per voxel.	Can suppress real sharp events if threshold is too low.

11.1 Practical decision tree

30. If an entire frame is corrupted globally, consider Frame Rejection or Scrubbing, not Despiking.
31. If abrupt temporal changes appear in global/DVARS traces but not every voxel is independently spiking, consider Scrubbing.
32. If isolated voxel/pixel spikes are sparse while the frame remains usable, consider Despiking.
33. If artifacts appear as slow spatial drift, consider Imregdemons instead.
34. If artifacts are frequency-specific and stable, consider Filtering, but only with a defined cutoff rationale.
35. Never stack Frame Rejection, Scrubbing, and Despiking automatically. Apply the least aggressive correction that solves the QC problem, then re-QC.

Why they are overlapping

Frame Rejection and Scrubbing both operate at the temporal-frame level and may identify similar bad frames. Despiking operates at the voxel-timepoint level. Running all three without checking can over-correct and create artificial smoothness or false FC.

12. Visualization: Time-Course Viewer, Video GUI, SCM GUI

12.1 Time-Course Viewer

Use the Time-Course Viewer to inspect raw and PSC dynamics, global/ROI traces, baseline behavior, stimulus timing, and preprocessing effects. In Standardized Analysis it is interactive; close the viewer when finished so the workflow can continue.

12.2 Video GUI

Video GUI displays dynamic fUSI data and overlays. Standardized launch defaults include baseline 30-35 s, signed PSC, caxis -100..100, and alpha modulation -20..20. Video GUI can also open group video bundles exported from Group Analysis, including multi-slice [Y X Z T] group data.

12.3 SCM GUI

SCM GUI computes and displays signal-change maps. The current code supports [Y X T] and [Y X Z T] PSC, baseline-window control, positive/negative/signed display modes, underlay handling, ROI exports, time-course PNG export, PowerPoint/series export, Video GUI launching, atlas warp, and SCM GroupAnalysis bundle export/reopening.

SCM setting	What it controls	Check
Baseline	Defines reference period for PSC.	Wrong baseline causes wrong maps.
Caxis	Controls color scaling of response values.	Saturation can hide differences or exaggerate noise.
Alpha modulation	Controls overlay transparency based on response magnitude.	Do not hide relevant negative/positive responses accidentally.
Underlay	Background image beneath the map.	A wrong underlay/slice can make a map look anatomically meaningful when it is not.
Mask	Restricts displayed/analyzed signal.	Bad masks create edge artifacts or exclude valid vessels.
Slice	Current Z slice for step-motor/multi-slice data.	Must match underlay, atlas label map, and ROI output.

12.4 SCM GroupAnalysis bundle export

36. Open SCM GUI on a validated subject dataset.
37. Confirm baseline, caxis, alpha modulation, polarity, underlay, mask, and slice.
38. If atlas-space group analysis is needed, use WARP FUNCTIONAL TO ATLAS first.
39. Export the full SCM GroupAnalysis bundle, not only a static PNG/map.
40. Load the full bundle into GroupAnalysis or reopen it in SCM/Video for verification.

13. Mask Editor

Mask Editor is used to create, load, and save masks and underlays. It supports brain/underlay masks and overlay/signal masks, and current defaults include a Standardized Doppler equalized underlay mode for clearer anatomical display.

- Use brain/underlay masks to restrict the anatomical background/valid brain area.
- Use overlay/signal masks to restrict where PSC/FC overlays are displayed or analyzed.
- Save mask bundles with clear dataset-specific names.

- For step-motor data, check masks slice-by-slice; do not assume a mask from one slice applies to all slices.
- If an underlay loads incorrectly in SCM or FC, verify the saved bundle fields and dimensions match the active dataset.

14. Atlas registration and segmentation

14.1 Registration to Atlas

Registration to Atlas routes users to the appropriate 2D or 3D workflow. For 2D probe and step-motor/multi-slice data, prefer the 2D coronal registration workflow unless the selected source is a true 3D anatomical volume.

File / workflow	Role
coreg.m	Unified atlas-registration launcher.
coreg_coronal_2d.m / registration_coronal_2d.m	Manual 2D coronal atlas registration for 2D and step-motor data.
coreg_3d.m / registration_ccf.m	Manual 3D atlas registration using atlas underlays.
interpolate3D.m	Resamples scan/anatomy into atlas-like voxel/grid context.
deConfUSlon_prepare_atlas.m / deConfUSlon_apply_rgb2acr.m	Prepare JM atlas color/order metadata; does not alter registration geometry.
Registration rule Atlas registration is a visual quality-control step. Check every slice. A wrong atlas transform propagates directly into segmentation, ROI labels, FC matrices, and group maps.	

14.2 Segmentation

In this toolbox, segmentation means atlas-label-based region-time-course extraction, not AI segmentation. Run Segmentation after atlas registration when ROI labels are needed.

- Confirm label maps match the active dataset dimensions and slice.
- For step-motor data, verify per-slice label maps and region-name files.
- Inspect region names, hemisphere signs/labels, and time courses before exporting to FC or GroupAnalysis.
- Do not trust segmentation if registration is poor or the wrong source slice was selected.

15. Functional Connectivity

FunctionalConnectivity.m is a large subject-level GUI for seed, ROI, pair, graph, and atlas-label FC. FC is sensitive to preprocessing, masks, global signal, motion, ROI labels, and slice selection.

15.1 Subject-level FC workflow

41. Choose the correct active dataset version; raw, filtered, scrubbed, PCA/ICA, and imregdemons outputs will not produce identical FC.
42. Load or verify a valid brain/signal mask.
43. Load ROI labels and region names that match the current functional space and slice.
44. For step-motor data, use recursive step-motor label/name discovery or the Use SCM atlas workflow.
45. Compute ROI current or seed maps and inspect heatmaps/time courses.
46. Check whether correlations are anatomically plausible or dominated by global/motion artifacts.
47. Export FC GroupAnalysis bundles only after the subject-level result is validated.

15.2 Step-motor and SCM atlas support

Current FC helper files support recursive discovery of step-motor label maps and region-name files, true slice-specific ROI FC calculation, JM atlas order/color support, and SCM atlas import. For step-motor data, sliceResults are critical because FC matrices and ROI overlays must represent the selected anatomical slice.

Field group	Why it matters
sliceIndex / sliceLabel	Identifies which step-motor slice the result belongs to.
ROI labels and names	Region IDs and readable labels; hemisphere signs should be preserved where appropriate.
ROI voxel counts	Number of voxels contributing to each ROI.
ROI mean time courses	Time courses used for ROI matrix calculation.
Pearson R and Fisher Z	Correlation matrix and z-space statistics matrix.
ROI atlas / label map / ROI map	Spatial map for overlays and slice-specific grouping.
subject metadata	Animal, scan/session, group/condition, source paths.

16. Group Analysis and FC Group Analysis

Group Analysis is the final cohort-level module. It should be used only after subject-level datasets, masks, atlas registration, segmentation, and bundles have passed QC.

16.1 Group Maps / SCM group workflow

48. Export full SCM GroupAnalysis bundles from SCM GUI for each validated subject.
49. Open Group Analysis and add/load subject rows.
50. Check Use rows, group labels, conditions, animal IDs, and scan/session labels.
51. Select the current step-motor slice where applicable.
52. Compute group map and inspect underlay, slice, polarity, caxis, and alpha modulation.
53. Export PNG/PPT/data only after the displayed map is correct.

16.2 Functional Connectivity Group Analysis

FC Group Analysis loads FC_GroupBundle_*.mat files exported from FunctionalConnectivity. Current logic is designed to attach FC bundles to existing rows rather than creating unintended new subject rows. This protects group metadata and preserves old Group Maps behavior.

FC-GA view	Purpose
Heatmap	Group-level ROI-by-ROI matrix; should reflect selected group/slice/hemisphere mode.
Subject Matrix	Subject-level matrix view.
Seed Profile	Seed-to-ROI correlation profile.
ROI Timecourse	ROI time-course comparison.
ROI Pair Summary	Selected pair statistics across animals/groups.
Animal Pair Values	Per-animal pair values; useful for outlier inspection.
ROI Overlay Map	Spatial map over underlay/atlas slice.
FC-GA safety rule After loading or attaching FC bundles, click Recompute FC Group and inspect the selected slice, ROI labels, hemisphere mode, and group assignment before any export.	

16.3 Hemisphere modes

FC and GroupAnalysis can represent left/right information in different ways. Merged mode combines left/right homologues; left-only and right-only restrict to one side; left-vs-right or both-separate modes preserve hemisphere-specific labels. A correct hemisphere mode should affect labels and, when the data differ between hemispheres, can affect computed matrices/maps. If left and right remain identical unexpectedly, verify atlas labels, segmentation outputs, FC export fields, and GroupAnalysis slice-specific data.

17. Outputs, naming, bundles, and logs

17.1 Typical output folders

Folder	Typical contents
QC	QC PNGs, MAT summaries, rejection traces, tSNR/CNR/common-mode outputs.
Preprocessing	New MAT datasets from Motor, imregdemons, scrubbing, filtering, smoothing, PCA/ICA, despiking.
PSC / Visualization	PSC data, SCM images, time-course PNGs, videos, PowerPoint exports.
Masks	Brain/underlay masks, overlay/signal masks, mask bundles.
Registration / Registration2D	Atlas transforms, Reg2D objects, atlas underlays, label maps.
Segmentation	Region-time CSV/MAT outputs, label maps, region tables.
FunctionalConnectivity	Seed maps, ROI matrices, heatmaps, graphs, FC GroupAnalysis bundles.
GroupAnalysis	Group maps, FC group summaries, PNG/CSV/Excel/PPT exports, group bundles.
Reports	Manual or generated reports and final documentation.

17.2 Dataset naming

The current code contains several deConfUSIon_* display/name helpers that preserve animal ID, scan/session ID, and processing-chain labels while using shorter physical filenames when paths become too long.

Good visible label	Bad / ambiguous label
1005_scan3_raw	dataset1
1005_scan3_motor_imregdemons	tmp_preproc_001
animal_scan_filter_band_0p01_0p20	filtered
1160_scan2_scm_bundle	Unknown_scm_output
Always export the Studio Log The Studio Log records analysis actions and warnings. Export or save it at the end of each analysis session so preprocessing decisions are reproducible.	

18. Publication-ready validation checklist

Area	Acceptance check
Raw data	Raw movie viewed; vessels/brain visible; no severe bubble/shadow/motion artifact.
TR and timing	TR, total time, stimulus/injection timing, and baseline window match acquisition log.
Active dataset	Dropdown points to the intended raw or processed version before every module.
QC	QC reviewed before and after preprocessing; failures resolved or documented.
Step-motor	Slice count/order and [Y X Z T] output verified.
Preprocessing	Every correction has a QC-based rationale; settings documented.
Frame handling	Rejected/scrubbed/despiked data amount is reasonable and not excessive.
Filtering/smoothing	Cutoffs/windows justified; effective TR updated where relevant.
SCM/Video	Baseline, caxis, alpha modulation, polarity, underlay, mask, and slice verified.
Masks	Brain/underlay and overlay/signal masks align with data.
Registration	Atlas overlay checked slice-by-slice.
Segmentation	ROI labels/names match anatomy and slice.
FC	Matrices, seed maps, ROI labels, hemisphere mode, and confounds checked.
Group Analysis	Subject rows, Use flags, groups, conditions, bundles, and slices verified.
Exports	Bundles reopened where possible; PNG/PPT/CSV/Excel checked manually.
Documentation	Studio Log, settings, excluded animals, and manual corrections saved.

19. Troubleshooting

Problem	Likely cause	Fix
Toolbox does not launch	Wrong folder/path or missing split GUI files.	Run from toolbox root; use deConfUSIon or run_fusi_studio; confirm fusi_studio_GUI.m and fusi_studio_callback.m exist.
Old behavior appears	Backup/archived file earlier on MATLAB path.	Remove backups from active path; run which functionName -all.
Dataset loads with wrong time	TR missing or incorrectly inferred.	Reload and manually confirm TR from acquisition log.
Standardized workflow stops	Interactive GUI remains open or a step failed.	Close the opened GUI to continue; read Standardized status and Studio Log.
SCM map looks wrong	Bad baseline, wrong active dataset, wrong underlay/mask/slice, motion artifact.	Check dropdown, baseline, raw movie, ROI trace, mask, and slice.
Filtered image looks dark	Mean not restored or old filtering file used.	Use current filtering.m; check restoreMean=true and path order.

Motor output wrong	Wrong slice count, frame grouping, or split filenames.	Inspect folder names, frames per slice, and motorInfo.
PCA/ICA removed response	Biological component selected as artifact.	Return to pre-PCA/ICA dataset and rerun with fewer/no removed components.
Segmentation labels wrong	Atlas transform or label map does not match functional slice.	Re-check registration, source slice, label dimensions, and region-name files.
FC heatmap globally high	Global signal/motion/non-brain voxels dominate.	Review QC, masks, time courses, and confound strategy.
FC GroupAnalysis bundle not attached	No selected/checked row or metadata mismatch.	Select/check intended rows before loading FC bundles; then Recompute FC Group.
PPT export fails	PowerPoint/ActiveX unavailable or path issue.	Check local Windows/Microsoft Office support and output path length.

20. Source-code orientation and release notes

20.1 Static source audit summary

The uploaded deConfUSlon.zip was statically inspected for this manual. The audit excludes .git and backups for runtime inventory and focuses on active source files. The active text/code set contains 88 files, including MATLAB source, README/manual text, atlas helper text, and optional acquisition support files.

Area	Files	Lines	Representative files
Launch / GUI / loader	10	16,382	deConfUSlon.m, run_fusi_studio.m, fusi_studio_GUI.m ...
Quality control	3	2,268	qc_fusi.m, frameRateQC.m, interpolateRejectedVolumes.m
Preprocessing	8	6,361	motor.m, imregdemons_preprocess.m, scrubbing.m ...
Visualization / PSC / masks	8	22,170	computePSC.m, fUSI_Live_Studio.m, SCM_gui.m ...
Registration / atlas / segmentation	13	10,106	coreg.m, coreg_coronal_2d.m, coreg_3d.m ...
Functional connectivity	9	12,632	FunctionalConnectivity.m, fc_export_group_analysis_bundle_interactive.m, deConfUSlon_FC_find_stepmotor_txt_names.m ...
Group analysis	5	20,800	GroupAnalysis.m, GroupAnalysis_Common.m, GroupAnalysis_FC.m ...
Dataset naming / compatibility	12	1,711	deConfUSlon_add_preproc_lazy_datasets.m, deConfUSlon_best_visible_dataset_name.m, deConfUSlon_canonical_dataset_label.m ...
UI helper / popup helper	12	1,044	deConfUSlon_fix_scm_video_dialog_fonts.m, deConfUSlon_force_fullscreen_fig.m, deConfUSlon_is_bad_display_name.m ...
Atlas tools / acquisition optional	6	7,184	atlas_tools/deConfUSlon_reorder_FC_by_list.m, atlas_tools/save_correct_colors.m, atlas_tools/list_selected_regions.txt ...

20.2 Files that must stay in the active release

Category	Files	Why keep
Core launch	deConfUSlon.m, run_fusi_studio.m, fusi_studio_GUI.m, fusi_studio_callback.m	Required for current split-GUI startup.
Data loading	loadFUSIData.m, loadDataCallback.m,	Loads data and detects TR/probe metadata.

	studio_load_options_dark_dialog.m	
Standardized workflow	standardizedAnalysis.m	Current Standardized Analysis GUI and Option A/B logic.
QC/preprocessing	qc_fusi.m, motor.m, imregdemons_preprocess.m, scrubbing.m, filtering.m, temporalsmoothing.m, pca_denoise.m, ica_denoise.m, despiking.m	Main analysis pipeline modules.
Visualization	computePSC.m, SCM_gui.m, play_fusi_video_final.m, fUSI_Live_Studio.m, mask.m	PSC, SCM, video, time course, mask workflows.
Atlas/segmentation	coreg*.m, registration*.m, Segmentation.m, atlas_tools/	Atlas registration and label extraction.
Functional connectivity	FunctionalConnectivity.m and deConfUSlon_FC_* helpers	Subject-level FC, step-motor labels, slice-specific FC, export.
Group analysis	GroupAnalysis*.m, tryReadSCMroiExportTxt.m	Group maps and FC group analysis.
Compatibility helpers	deConfUSlon_*display/name/path/popup helpers	Preserve readable dataset names, path-safe outputs, GUI layout behavior.

20.3 Release package recommendation

- Ship a clean ZIP containing active runtime files, atlas_tools, required atlas resources or download instructions, README, and this manual.
- Exclude .git, backups, generated QC/Preprocessing/PSC/Visualization/Registration/GroupAnalysis outputs, and old patch archives from user release ZIPs.
- Keep backups outside the release package after validation, but do not delete historical backups until representative 2D and step-motor testing passes.
- Keep metadata compatibility helpers even if names contain legacy HUMOR fields, because old outputs may depend on them.

Appendix A. Formula quick reference

Quantity	Formula	Use
Baseline mean	$B = \text{mean}(I(t))$ over baseline frames	Reference intensity for PSC.
PSC	$\text{PSC} = 100 * (I - B) / B$	Main signal-change quantity.
Effective TR after interpolation	$\text{TR}_{\text{eff}} = \text{TR} / N$	Used if PSC temporal interpolation factor N is applied.
Nyquist	$f_N = 1 / (2 * \text{TR})$	Maximum resolvable temporal frequency.
Smoothing window frames	$\text{winVol} = \max(1, \text{round}(\text{winSec} / \text{TR}))$	Converts seconds to frames.
Block subsampling TR	$\text{TR}_{\text{out}} = \text{TR} * \text{winVol}$	Required when block/subsample mode shortens data.
tSNR	$\text{mean}(\text{signal}) / \text{std}(\text{signal})$	Temporal stability/noise measure.
DVARs	$\text{RMS}(I(t) - I(t-1))$ over voxels	Framewise temporal outlier metric.
Pearson r	$\text{corr}(x, y)$	FC between two time courses.
Fisher z	$z = \text{atanh}(r)$	Use for averaging/testing correlations.
Back-transform	$r = \tanh(z)$	Use for display after z-space averaging.

Appendix B. Active source-file inventory

This appendix lists the current active files by functional category. Backups and .git content are intentionally excluded.

Launch / GUI / loader

File	Lines
deConfUSlon.m	16
run_fusi_studio.m	154
fusi_studio_GUI.m	6,411
fusi_studio_callback.m	6,050
loadFUSIData.m	921
loadDataCallback.m	294
datasetDropdownCallback.m	33
studio_load_options_dark_dialog.m	570

standardizedAnalysis.m	910
standardizedAnalysis_v2.m	1,023

Quality control

File	Lines
qc_fusi.m	2,062
frameRateQC.m	165
interpolateRejectedVolumes.m	41

Preprocessing

File	Lines
motor.m	1,741
imregdemons_preprocess.m	509
scrubbing.m	588
filtering.m	562
temporalsmoothing.m	270
pca_denoise.m	1,049
ica_denoise.m	1,511
despike.m	131

Visualization / PSC / masks

File	Lines
computePSC.m	268
fUSI_Live_Studio.m	2,438
SCM_gui.m	7,906
play_fusi_video_final.m	6,421
mask.m	3,542
showScmVideoSetupDialog.m	1,024
blackbdy_iso.m	286
winter_brain_fsl.m	285

Registration / atlas / segmentation

File	Lines
coreg.m	851
coreg_coronal_2d.m	1,454
coreg_3d.m	794
registration_coronal_2d.m	1,783
registration_ccf.m	2,015
interpolate3D.m	90
mapscan.m	109
moveimage.m	285
Segmentation.m	2,187
readFileList.m	133
deConfUSlon_prepare_atlas.m	56
deConfUSlon_apply_rgb2acr.m	144
deConfUSlon_find_stepmotor_seg_fc_files.m	205

Functional connectivity

File	Lines
FunctionalConnectivity.m	10,701
fc_export_group_analysis_bundle_interactive.m	372

deConfUSlon_FC_find_stepmotor_txt_names.m	236
deConfUSlon_FC_force_layout.m	308
deConfUSlon_FC_make_slice_roi_result.m	301
deConfUSlon_FC_read_region_names_file.m	240
deConfUSlon_FC_remember_layout.m	128
deConfUSlon_FC_stepmotor_read_folder.m	262
deConfUSlon_fc_jm_order.m	84

Group analysis

File	Lines
GroupAnalysis.m	13,956
GroupAnalysis_Common.m	5,019
GroupAnalysis_FC.m	362
GroupAnalysis_Map.m	1,415
tryReadSCMroiExportTxt.m	48

Dataset naming / compatibility

File	Lines
deConfUSlon_add_preproc_lazy_datasets.m	282
deConfUSlon_best_visible_dataset_name.m	46
deConfUSlon_canonical_dataset_label.m	399
deConfUSlon_commit_full_display_name.m	18
deConfUSlon_compact_chain_name.m	39
deConfUSlon_display_from_file_context.m	50
deConfUSlon_display_name_from_sources.m	361
deConfUSlon_fix_processing_name.m	300
deConfUSlon_fix_studio_dataset_names.m	43
deConfUSlon_make_loaded_display_name.m	25
deConfUSlon_safe_preproc_save_path.m	119
deConfUSlon_write_full_display_metadata.m	29

UI helper / popup helper

File	Lines
deConfUSlon_fix_scm_video_dialog_fonts.m	157
deConfUSlon_force_fullscreen_fig.m	46
deConfUSlon_is_bad_display_name.m	13
deConfUSlon_pcaica_scope_tag.m	10
deConfUSlon_popup_autofit_apply.m	10
deConfUSlon_popup_autofit_timer.m	89
deConfUSlon_popup_polish_now.m	392
shortenMiddle.m	54
studio_mkdir.m	6
studio_resolve_paths.m	175
buildFooterLabel.m	7
addLog.m	85

Atlas tools / acquisition optional

File	Lines
atlas_tools/deConfUSlon_reorder_FC_by_list.m	20
atlas_tools/save_correct_colors.m	36
atlas_tools/list_selected_regions.txt	290
vfUSI_StimBox_TTL_EACH_FRAME_OR_TRIGGER_ACCESSORIES_COMMAND.m	3,371

vfUSI_StimBox_TTL_EACH_FRAME_OR_TRIGGER_ACCESSORIES_OBJECT.m	775
vfUSI_StimBox_TTL_EACH_FRAME_OR_TRIGGER_ACCESSORIES_PARAMETERS.m	2,692

Appendix C. Change log for this manual update

Date	Change
2026-06-24	Merged the 09.06.2026 manual structure with the 24.06.2026 Standardized Analysis update.
2026-06-24	Inspected the uploaded current deConfUSlon.zip source for active files, line counts, module roles, and standardized workflow code.
2026-06-24	Added expanded abbreviations and key terms.
2026-06-24	Added mathematical principles for PSC, effective TR, filtering, smoothing, QC metrics, and FC/Fisher z.
2026-06-24	Expanded Workflow A Fast and Workflow B Detailed, including ticked/unticked steps and rationale.
2026-06-24	Added instructions for editing row order, checking/unchecking Run, editing numeric presets, running a single step, running the workflow, and saving/loading presets.
2026-06-24	Added a dedicated comparison of Frame Rejection, Scrubbing, and Despike, emphasizing their overlap and differences.
2026-06-24	Updated SCM, Video GUI, Mask Editor, atlas registration, segmentation, FC, and GroupAnalysis explanations with step-motor and bundle-specific cautions.
2026-06-24	Added publication-ready checklist, troubleshooting, source-code orientation, and release package guidance.